RHCA White Test

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1 Networking Configuration

First, configure your Host Name, IP Address, Gateway and DNS. Then start the exam.

1.1 Machine 1

- IP Address: `192.168.1.50/24`
- Gateway: `192.168.1.1`
- DNS: `192.168.1.1`
- Hostname: `client.example.com`

1.2 Machine 2

- IP Address: `192.168.1.30/24`
- Gateway: `192.168.1.1`
- DNS: `192.168.1.1`
- Hostname: `server.example.com`

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2 Machine 1 Tasks

1. **Set multi-user target as default**
   - Set multi-user target as your default target. Then, reboot. you should begian your work in the multi-user target y.
2. **Find lines containing "seismic"**
   - Find all lines in the file `/usr/share/dict/words` that contain the string `seismic`. Put a copy of all these lines in the file `/root/wordlist`.
3. **Create user David with password aging**
   - Create a user `David` with password `tekup`. Configure password aging for david: Set the mindays to 7, maxdays to 28, and warndays to 5.

4. **Create groups with shared GID**
   - Create a group called `linuxadm` with GID 5000 and another group called `dba` sharing the GID 5000. Add David as a secondary member to group `linuxadm`.
5. **Modify group name and GID**
   - Change the `linuxadm` group name to `sysadm` and the GID to 6000. Modify the primary group for David to `sysadm`.
6. **Create archive directory**
   - Create a directory `/archive` so that only the owner and the group owner can fully access it. Under `/archive`, create a file named `targets` that contains the list of targets.
     Note: use systemctl list-unit-files –type=target to display all targets
7. **Create user melinda**
   - Create the user `melinda` with uid 4525 with password `redhat` and gid 3200.
8. **Configure default permissions for melinda**
   - All files that will be created by melinda should have permissions 664 and all directories should have permissions 764.
9. **Copy archive with specific permissions**
   - Copy recursively `/archive` to `/root/archive2`. The owner of `/root/archive2` is david. Its group is sysadm. All files that will be created in `/root/archive2` belong to sysadm group.
10. **Create a services file with specific permissions**
    - Copy the list of services to a file named `services` under the `/srv/admin` directory.
    - Permissions:
      - User `melinda` can read and delete the file.
      - Members of the group `sysadm` can access the file.

      Note: Assuming the list of services is output from systemctl list-units –type=service
11. **Configure YUM repositories using the provided links**

    Note: For the sake of this test to function correctly, do not modify anything under `/etc/yum.repos.d`. Instead, create two files named `BaseOS.repo` and `AppStream.repo` under the `/root` directory.

    - Repositories: BaseOS and AppStream
    - BaseOS URL: `http://content.example.com/rhel9.0/x86_64/dvd/BaseOS`
    - AppStream URL: `http://content.example.com/rhel9.0/x86_64/dvd/AppStream`

12. Create backup archive

- Create a backup file named `/root/backup.tar.bz2` which contains the content of `/usr/local`.

13. Configure AutoFS

- Automount NFS home directories for `ldapuserX` at `/home/guests/ldapuserX`
- NFS Server: `classroom.example.com (172.25.250.254)`
- Ensure login works for users `ldapuser1` to `ldapuser20`

Note: For the sake of this test to function correctly, Write your answers in a file named `/root/AutoFS` using the following format with an absolute path like (`/home/user/dir/file`) :

```
----------------------------------------------------
First file to modify : <name of the file>
Content of the first file : <insert content here>

Second file to modify : <name of the file>
Content of the second file : <insert content here>

The Command you execute after all of this : <insert command here>
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```

14. Create and run a PDF converter container

- Create a user named `athena` with password `trootent` and install necessary container tools.

- Create directories `/data/input` and `/data/output` with proper ownership and permissions for the athena user (full access).

- Download the Python script and Dockerfile from

  -https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/master/pdf_converter.py
  -https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/master/Dockerfile
- Build a container image named `pdf` and run a container named `pdfconverter` with volume mappings from `/data/input` to `/data/input` and `/data/output` to `/data/output` inside the container.

- Create a test file `file.txt` in the input directory and verify the PDF conversion works.

15. Configure systemd service for the PDF converter container

- As the `athena` user, create a systemd user service for the existing `pdfconverter` container.

- The service should be named `container-pdfconverter.service` and must be enabled to start automatically.

- Configure user linger for the athena user to ensure the service runs even when the user is not logged in.

- Verify that the service is active and will restart automatically on system reboot.

16. Configure HTTP web server

- Configure a http web server which can be accessed through port 87.

17. Write file search script

- Write a script that finds all the files owned by david and have size greater than 10KB and less than 10M and store them in `/tmp/find1`.

18. Configure NTP client

- Configure this machine as a client NTP. The server is `africa.pool.ntp.org`.

3 Machine 2 Tasks

1. **Reset root password**
   - Password: `trootent` via specific boot procedure
2. **Configure password policy**
   - All passwords should contain digits, uppercases and lowercases. The minimum acceptable size for new passwords is 7.
3. **Write a backup script for small files**
   - Create a shell script named `backup-smallfiles.sh` under the `/root` directory.
   - The script should search for all files under `/usr` that are smaller than 2MB and copy them to `/root/backup`.
4. **Schedule the etc backup script**
   - As a System Administrator, you are responsible for backing up the `/etc` directory every night.
   - Write a shell script named `/root/backup-etc.sh` that uses the `tar` command to back up the `/etc` directory.
   - Schedule this script to run every night at **11:00 PM**, excluding Sundays.
5. **Create swap partition**
   - Create a 2G swap partition which take effect automatically at boot-start and should not affect the swap partition.
6. **Create volume group and logical volumes**
   - Create a volume group called `vgfs` of 160MB created on the `/dev/sdb` disk. The PE size for the volume group should be set at 16MB. Create 2 logical volumes called `ext4vol` and `xfsvol` of size 80MB each and initialise them with the Ext4 and XFS file system types. Ensure that both file systems are persistently defined using their logical volume device filenames. Create mount points `/ext4vol` and `/xfsvol`, mount the file systems, and verify their availability and usage.
7. **Extend logical volume**
   - Extend the `xfsvol` logical volume along with the file system it contains. The new size should be between 157M and 175M.
8. **Set tuned profile**
   - Set recommended profile as your tuned profile.

Good Luck!